When AI Cannot Understand Xiaolongbao

From working in a restaurant to developing an app, a high school student asks whether the convenience technology promises reflects its users' languages and circumstances

By Catherine Lin



During the lunch rush at the Chinese restaurant where I worked, I was busy packing three takeout orders when a customer on the phone asked me to repeat the beef dishes on the menu. I could not stop packing, but I could not leave the phone unanswered either. I began to wonder: Could a machine help handle calls and take orders?

It was the summer before my senior year of high school. At the time, the answer seemed straightforward: introduce an AI phone-ordering system. But when I brought the idea to my boss, the response surprised me. "We've tried that." I volunteered to test it myself. Having grown up with technology, I felt that perhaps I could find a way to make the system work smoothly.

"I'd like an order of xiaolongbao," I said into the phone. "Sorry, I didn't understand." I slowed down and tried again, but the system still could not recognize the dish. Only when I switched to the familiar English description, "soup dumplings," did it finally identify the item and ask me to confirm my order. The exchange was brief, but it unsettled my initial assumptions. I had thought the restaurant was the one that needed help: it needed to keep up with technology. Now I began to ask whether technology had kept up with the restaurant's actual needs. One failed order is certainly not enough to judge AI's performance across the restaurant industry. But it gave me a question worth pursuing: When restaurant technology promises convenience, whose language, whose menu, and whose working conditions does that promise take into account?

That summer, I began interviewing Asian restaurant operators to learn which tools they used, where technology helped, and where it fell short. Those conversations also became the starting point for BridgeBite, an app I began developing to help restaurants with tasks such as translation, marketing, and cost tracking. The interviews, however, showed me that the situation was far more complicated than I had imagined.

Asian restaurants' experiences with AI varied. Some had tested phone-ordering tools and found them unreliable. Others were still discussing whether to adopt them. Still others were already using AI for marketing, while relying on staff to check work the software could not complete on its own. Those differences mattered. If I wanted to build a useful tool, I needed to understand them before deciding on a solution.

Ordering Systems Must Withstand the Lunch Rush

In an interview conducted in Chinese at Fang's Bistro, owner Jack Fang repeatedly described his experience trying AI phone ordering. The problems included misunderstandings, delayed responses, and difficulty handling accents or special requests. A customer might want a vegetable left out or an extra portion added. For someone taking orders, these are routine requests. For a system that cannot reliably understand them, they can lead to incorrect orders and the need for human intervention. The interview also challenged another common assumption: that restaurants avoid new technology mainly because it costs too much. Fang said cost was not the central issue. What mattered was whether the technology was reliable enough for the restaurant to depend on. That distinction changed my understanding of successful adoption. A system might answer the phone and recognize dish names yet still get the details customers care about most wrong. When evaluating a tool, restaurants have to weigh the cost of those errors against the time it might save.

This was not a wholesale rejection of AI. Fang looked forward to better phone-ordering technology and saw potential in marketing and other tools that could reduce the workload and attract customers. What frustrated him was the gap between technology's promises and the experience of actually using it.

Pete Immanivong, manager of Thaicoon Restaurant & Pub, described a different situation. The management team had discussed possible uses for AI, such as answering questions about opening hours and working with food images, but had not adopted it at the time of the interview. "We've only discussed it," he said. The restaurant already had a way to present its food: a professional photographer visited several times a year to capture dishes and changes in plating. New technology would be entering a restaurant with an established workflow. It needed to give the operators a compelling reason to change how they already worked.

AI Can Help but Still Needs Human Oversight

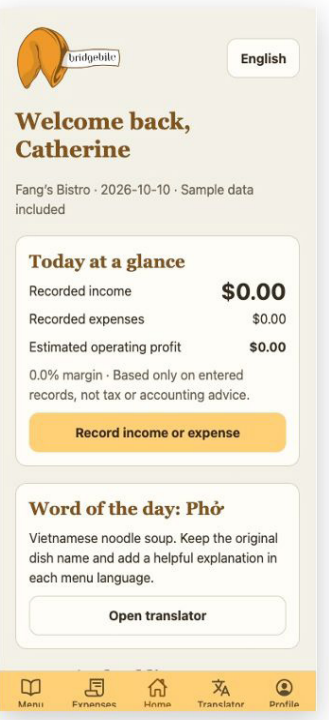
Gabriela Torres, the manager of a Korean fried chicken restaurant, offered another perspective. She was already using AI to help create marketing materials, brainstorm menu content, and write social media posts. She said these tools helped reduce design costs. Her experience showed why such tools appeal to restaurant managers. Making a poster or finding the right wording for a social media post takes time, even when the finished product looks simple. Having a draft makes it easier to get started.

Yet a tool's usefulness also depends on whether users know when to ask someone else for help. Torres does not speak Korean. When AI generates Korean text for a poster, she asks someone who knows the language to review it. "I ask him to double-check it every time," she said. The software can generate text easily, but she cannot assess its accuracy just as easily. Before the restaurant puts that content in front of customers, someone with the necessary language skills still needs to check it. Torres also wanted more convenient tools for improving food photos, such as removing a cluttered kitchen background. Although the restaurant uses professional photography services, staff still have to handle occasional photo edits between shoots.

A follow-up interview at Fang's Bistro returned to a related issue: maintaining the restaurant's visibility across online platforms takes time and effort. The interviewee wanted to expand promotional efforts but lacked the capacity to manage additional channels. These interviews shifted my attention away from simply asking whether a restaurant had "adopted AI." A more meaningful question was what actually changed after it began using a tool. Did the tool eliminate a task, make the work easier, or merely move the manager's error-correction work somewhere else?

Whose Language Can Technology Understand

As an Asian American student, I kept returning to that moment when the ordering system understood an English description but failed to recognize the Chinese name of a dish. Swwitching to "soup dumplings" did allow me to place the order. But the question remained: How much do people have to change the way they speak for a tool to understand them?



Existing research reminds us that AI's representation of food and culture deserves close examination. In 2024, Stanford researchers Yiwei Luo, Kristina Gligorić, and Dan Jurafsky analyzed restaurant reviews in the United States alongside reviews generated by GPT-3.5. They found that some reviews portrayed immigrant cuisines as foreign or lower in status, including descriptions that framed Asian and Latin American cuisines as "exotic." AI-generated reviews reproduced some of these patterns. The study did not test the ordering system I used, nor can it establish how all current AI products treat Asian restaurants. It does, however, show that discussions of a system's usefulness should also consider how it understands and represents culture.

Likewise, my interviews were only a handful of conversations, not a representative survey. They cannot prove that technology companies deliberately overlook Asian restaurants, or that every difficulty the interviewees described is unique to those restaurants. Limited time, marketing costs, and unreliable software are challenges many small businesses may face. Still, even a widespread problem requires understanding the people experiencing it. For the restaurants I interviewed, useful technology needs to account for specific languages, dishes, customer requests, and the limited time available to staff. If users must continually correct a tool's output, simply calling the product easy to use does not mean much.

Listen First Then Build

The BridgeBite app is my attempt to respond to these interviews. Features currently in development include translation, assistance with social media and advertising content, cost tracking, and AI-assisted business advice. I am also exploring ways for operators to edit a menu in one place and have those changes update across platforms such as Uber Eats and DoorDash. That integration remains a development goal.

I have shown the app to restaurants, but none has yet tested it in day-to-day operations. I therefore cannot claim that it has saved managers time, increased revenue, or solved the problems raised in the interviews. Those questions need to be answered through testing.

The interviews have also led me to set more demanding standards for those tests. Translations need to be reviewed by people who know the relevant languages. Marketing suggestions must suit the restaurant using them. Menu tools need to handle a range of requests and changes, because a real-world menu is far more complicated than rows of dish names on a screen.

These conversations also changed how I understood my own role. I began with confidence that I could bring better technology to restaurants. Only after listening did I recognize how much knowledge the operators had already accumulated: which tools they had tried, which alternatives worked, and which tasks they would gladly hand over if they found a reliable tool.

Today, I am a senior at Bridgeland High School, still developing the app and still learning what questions to ask. BridgeBite gives me a chance to turn those questions into action. Whether the answers are useful is something the restaurants must help determine.